

popquiz kinematics 1D -1

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:26pm

Quiz Instructions

1 attempt



Question 1 15 pts

A postal truck driver driving due east gently steps on her brake as she approaches an intersection to reduce the speed of the truck. What is the direction of the truck's acceleration, if any?

(hint: the acceleration has the same direction as the force)



due south



due west



due east



there is no acceleration



Question 2 15 pts

mcats

A car is traveling at $40 \frac{\text{km}}{\text{hr}}$ and the driver puts on the brakes, bringing the car to rest in a time of 6 s. What is the magnitude of the average acceleration of the car?

hint: 1 hour=3600s



24,000 km/hr/hr



240km/hr/hr



30,000 km/hr/hr



12,000 km/hr/hr



Question 3 17 pts

Molly (my gecko) walks 3 inches @ east then 5 inches @ west. Her displacement is about



2 inches @ east



2 inches @ west



8 inches @ east



8 inches @ west



Question 4 10 pts

A mosquito flying at 3 m/s that encounters a breeze blowing at 3 m/s in the opposite direction has a speed of (relative to Gnd)

hint: swimming counter current.





6 m/s



4 m/s



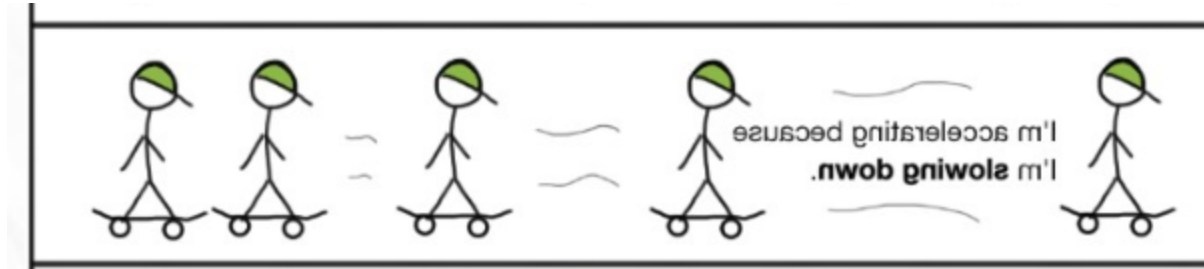
0



3m/s



Question 5 17 pts



The skateboarder is moving along a number line. + is @right and - is @ left.

What is true (**he is moving to the left ! <-----**)



his velocity is negative



We can't tell. Depends on the color of the hat



his velocity is positive



his acceleration is negative



his acceleration is 0



Question 6 17 pts

If a rocket initially at rest accelerates at a rate of 50 m/s/s for one minute, its speed will be



3000 m/s.



3600 m/s.



50 m/s.



500 m/s.



Question 7 4 pts

Molly (my gecko) leaves its hide and walks 5 inches to grab a cricket. She then go back to her hide. It takes 5s for the ROUND TRIP. ROUND TRIP/ Her average velocity is

☐

5 inches/s

☐

10 in/s

☐

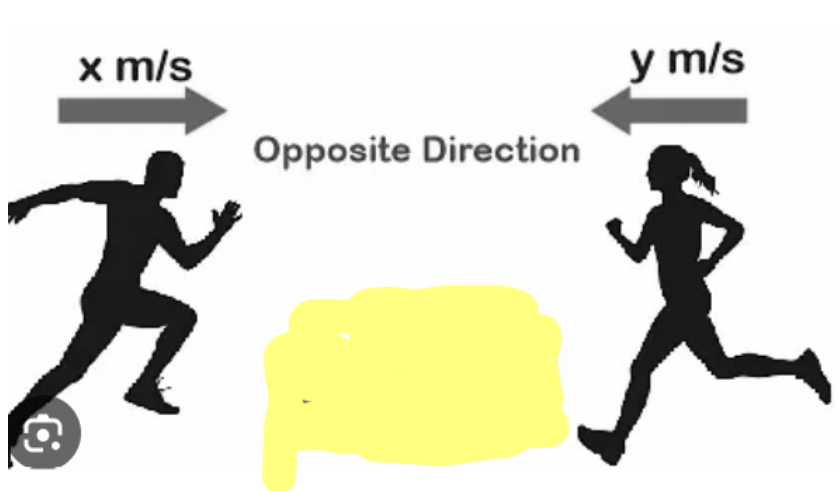
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☐

1 in/s

☐

Question 8 4 pts



$x=3\text{m/s}$ and $y=2\text{m/s}$

Their relative speed is (don't worry about sign)

☐

2m/s

☐

1m/s

☐

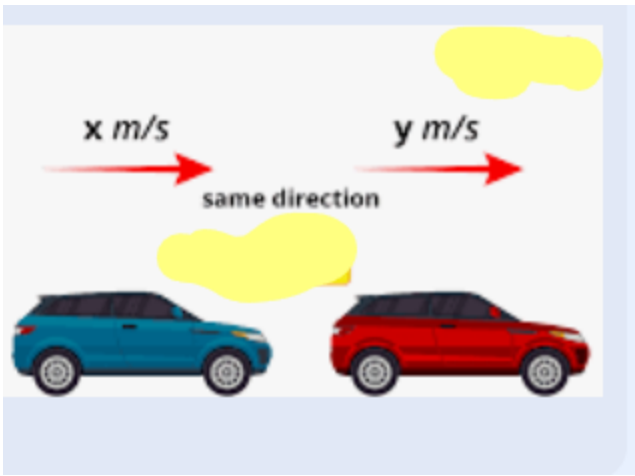
5m/s

☐

3m/s

☐

Question 9 4 pts



$x=10\text{m/s}$ $y= 15\text{m/s}$

☐

relative speed is 10m/s

☐

relative speed is 5m/s

☐

relative speed is 0

☐

relative speed is 25m/s

⋮

Question 10 2 pts

bonus

the slope of a graph $x(t)$ position as a function of time. At a given time is the

☐

displacement

☐

average velocity

☐

average acceleration

☐

instaneous velocity

Not saved

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